

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE FOR VEHICLE

International[®] Electric MV[™] Series

Truck (eMV[™])

Model Year 2021–Present

Battery Electric Vehicle



0000488712

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0. Rescue Sheet



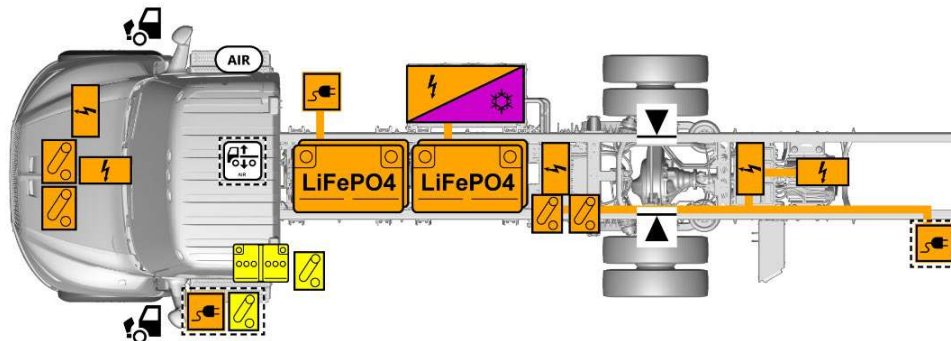
International® Electric MV™ Series Truck (eMV™) Model Year 2021–Present



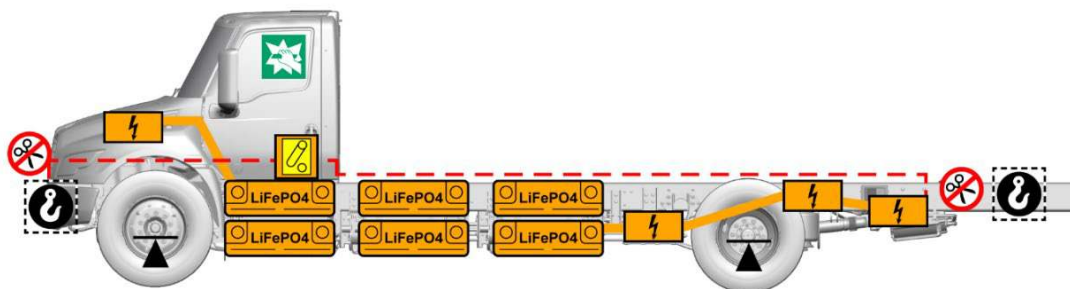
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690 V



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WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, NEVER CUT OR ATTEMPT TO BREACH HIGH-VOLTAGE BATTERIES, HIGH-VOLTAGE COMPONENTS, OR HIGH-VOLTAGE WIRING.

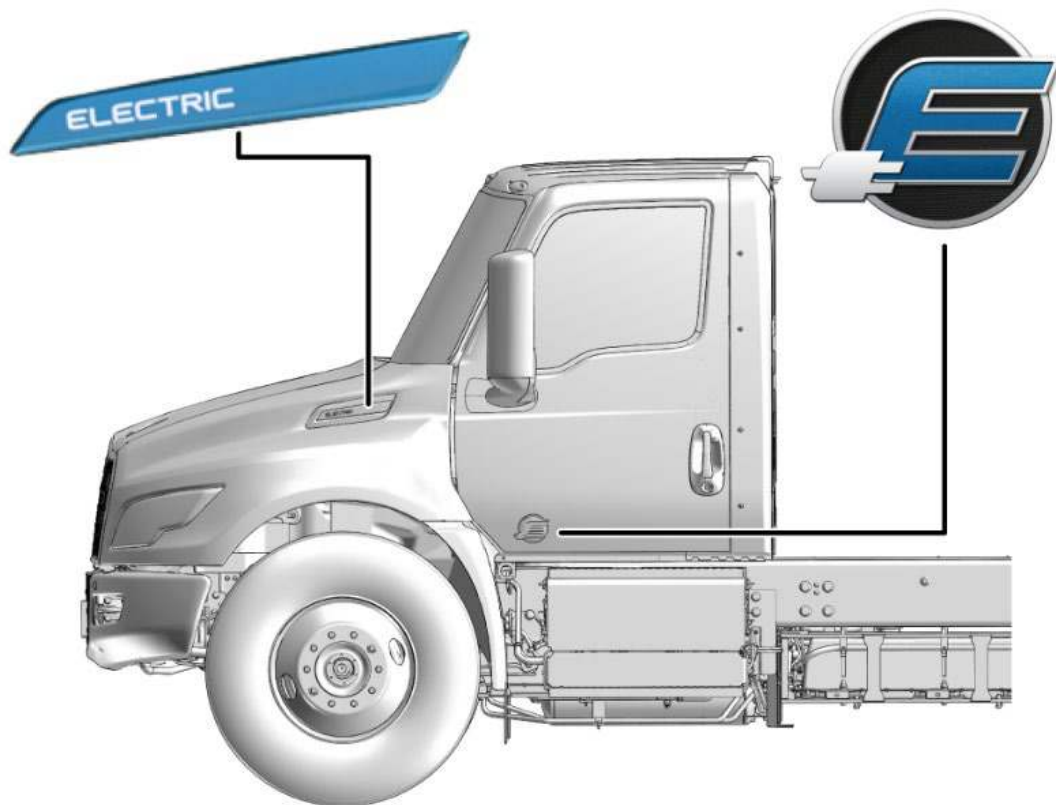
	Low-Voltage Battery		High-Voltage Battery		High-Voltage Component		High-Voltage Charge Port		Battery Thermal Mgmt. System
	Low-Voltage Disconnect		12V Shutoff for High-Voltage		High-Voltage Disconnect		Optional Tow Hook		High-Voltage Power Cable
	Lift Point		Air Tank		Break To Obtain Access		Hood Release		No Cut Below Line
	Height Control Air		Optional Feature		Electric Propulsion				

1. Identification / Recognition

Exterior Identification

The following markings identify the vehicle as electric powered. Some localities may not include all badging or decals.

- BLUE ELECTRIC decal on both sides of the hood
- Badges on left and right front doors



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Figure 1. Exterior Identification

1. Identification / Recognition

Interior Identification



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Figure 2. Instrument Cluster

- 1. REGEN
- 2. Regenerative Braking System (RBS) indicator, Levels 1, 2, 3

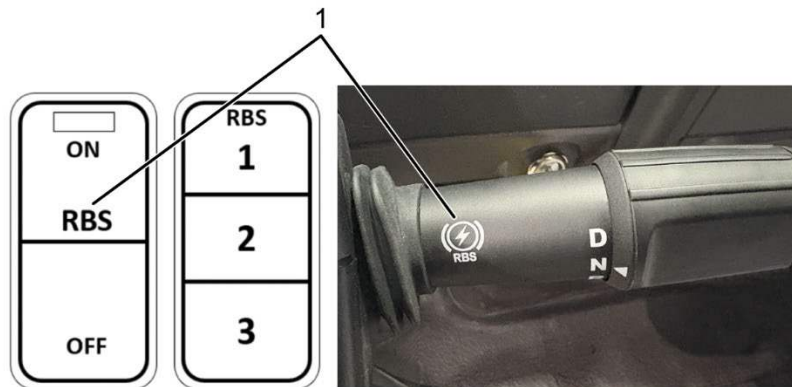
The word REGEN (Figure 2, Item 1) and Regenerative Braking System (RBS) indicator, levels 1, 2, 3 (Figure 2, Item 2) are on the bottom of the instrument cluster.

Icon	Name
 0000490483	REGEN
 0000488716	Regenerative Braking System (RBS) indicator, Levels 1, 2, 3

1. Identification / Recognition

Interior Identification

A Regenerative Braking System (RBS) will be optioned with dash-mounted switches or a steering column right-side stalk.



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Figure 3. Regenerative Braking System (RBS)

1. RBS

High-Voltage Component Identification



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Figure 4. High-Voltage Component Identification

High-voltage cables and components can be identified by the following:

- A triangle with a lightning bolt
- ORANGE in color and / or ORANGE with stripes

2. Immobilization / Stabilization / Lifting

Immobilization

When approaching the truck, check if the truck is ON by the following indications:

- The headlights, tail lights, and marker lights are ON.
- A humming sound will also come from under the hood.
- Front and rear external speakers produce an audible tone.

NOTE: When the vehicle is shut down for any reason, the motor returns to Neutral. The vehicle can roll if the parking brake is not set.

Immobilization or stabilization of a compromised vehicle can be achieved by the following methods:

1. Turn OFF vehicle and set parking brake (Refer to Section 3 – Disable Direct Hazards / Safety Regulations).
2. Insert an emergency plug into the charge port. It will only immobilize the vehicle but will not disconnect the high-voltage system.
3. Install wheel chocks.

NOTE: An emergency plug only immobilizes the vehicle by automatically engaging the parking brake and setting the motor to Neutral. The emergency plug does not disconnect power from the high-voltage circuits.

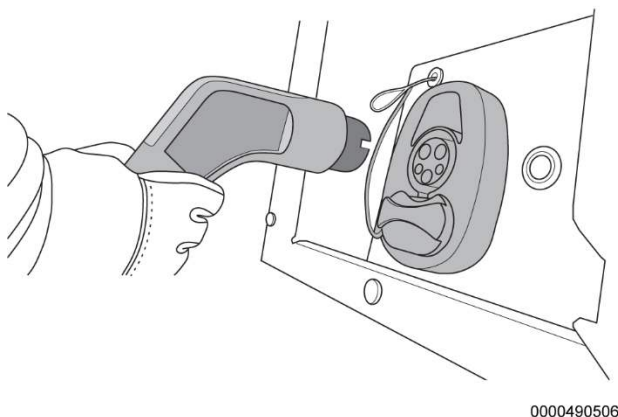


Figure 5. Emergency Plug

2. Immobilization / Stabilization / Lifting

Lifting

Do not lift or apply force under high-voltage batteries, components, or cables. Wheel lifts or platform lifts are recommended.

In a roadside condition when these lifts are not available, raise the vehicle under the axles at the lifting points shown in Figure 7. Install support stands under the axles and side members as needed for stability.

If equipped, use the tow hooks to lift or recover the vehicle short distances as needed.

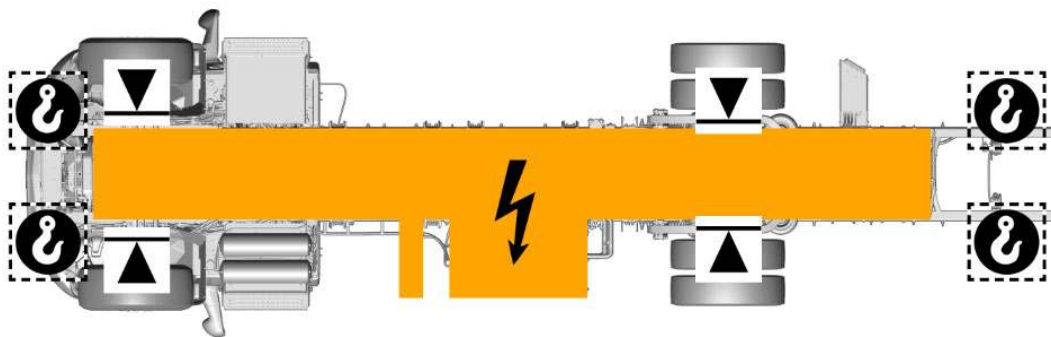
The in-cab dash-mounted switch will lower the rear height of the vehicle by releasing the air in the rear air springs.



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Figure 6. Switch

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, OR DAMAGE TO PROPERTY, DRAIN ALL AIR FROM THE SUSPENSION SYSTEM PRIOR TO MOVING A COMPROMISED VEHICLE WHENEVER POSSIBLE. FAILURE TO DO SO COULD RESULT IN THE VEHICLE UNCONTROLLABLY SHIFTING OR MOVING DUE TO AIR BEING RELEASED.



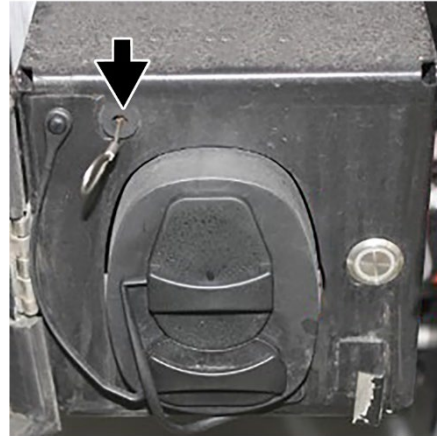
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Figure 7. Lifting Points and Tow Hooks

3. Disable Direct Hazards / Safety Regulations

Emergency Charger Release

To manually release a locked charger and stop the charging process, pull the emergency charger plug release cable.

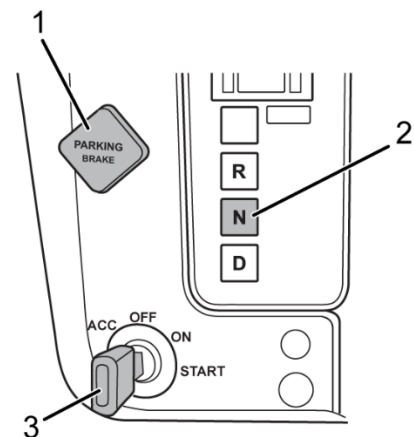


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Figure 8. Emergency Charger Plug Release Cable

Turning OFF the Vehicle

1. Push Neutral (N) button (Figure 9, Item 2) or turn stalk shifter to Neutral (N) position (Figure 10, Item 1).
2. Set the parking brake by pulling the parking brake knob (Figure 9, Item 1).
3. Turn the key (Figure 9, Item 3) to the left to turn OFF the vehicle.
4. Remove the key.



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Figure 10. Stalk Shifter

1. Neutral

3. Disable Direct Hazards / Safety Regulations

Turning OFF 12V and High-Voltage Disconnect Switches

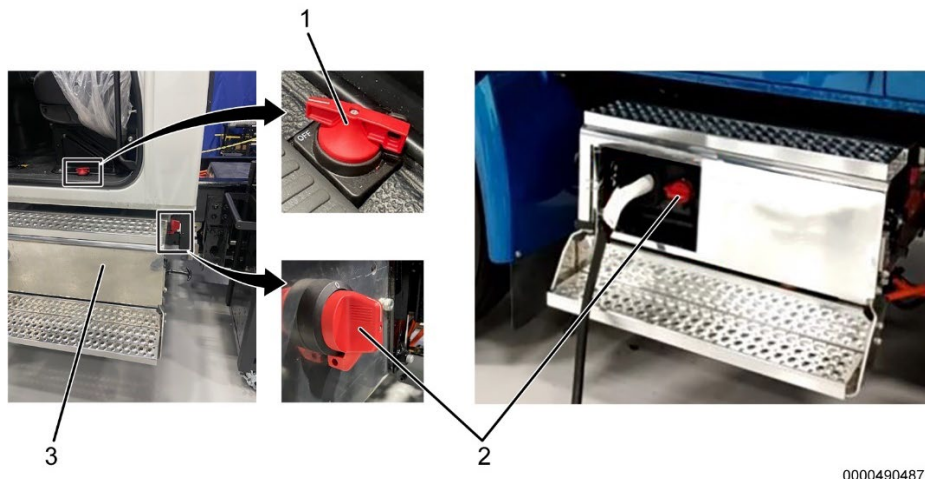


Figure 11. 12V and High-Voltage Disconnect Switches

1. High-voltage disconnect switch (located inside of cab)
2. 12V disconnect switch
3. 12V battery compartment

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS REMEMBER THE HIGH-VOLTAGE DISCONNECT SWITCH ONLY ISOLATES HIGH-VOLTAGE FROM COMPONENTS AFTER THE S-BOX. THE HIGH-VOLTAGE DISCONNECT SWITCH DOES NOT DISCONNECT HIGH-VOLTAGE IN THE S-BOX, THE HIGH-VOLTAGE BATTERIES, OR THE CABLES BETWEEN THESE ITEMS. THE VEHICLE'S 12 VOLT SYSTEM COULD STILL BE ENERGIZED. ALWAYS SHUT OFF BOTH THE HIGH-VOLTAGE DISCONNECT SWITCH AND THE 12V DISCONNECT SWITCH WHENEVER INTERACTING WITH A DAMAGED VEHICLE. NEVER TOUCH ORANGE CABLES WITHOUT WEARING APPROPRIATE HIGH-VOLTAGE PERSONAL PROTECTION EQUIPMENT (PPE).

1. Open the driver-side door.
2. Turn ignition key to OFF position.
3. Turn the RED 12V disconnect switch (Figure 11, Item 2) to OFF position.
4. Turn the RED high-voltage disconnect switch (Figure 11, Item 1) to OFF position.
5. Wait 3 minutes for high-voltage energy to dissipate.

NOTE: Components in high-voltage zone 2 (BLUE-striped cables) and high-voltage zone 3 (WHITE-striped cables) remain live after the high-voltage disconnect switch is turned OFF. These zones are located primarily between the frame rails on the underside of the vehicle and are labeled.

3. Disable Direct Hazards / Safety Regulations

Disconnecting Manual Service Disconnects (MSDs)

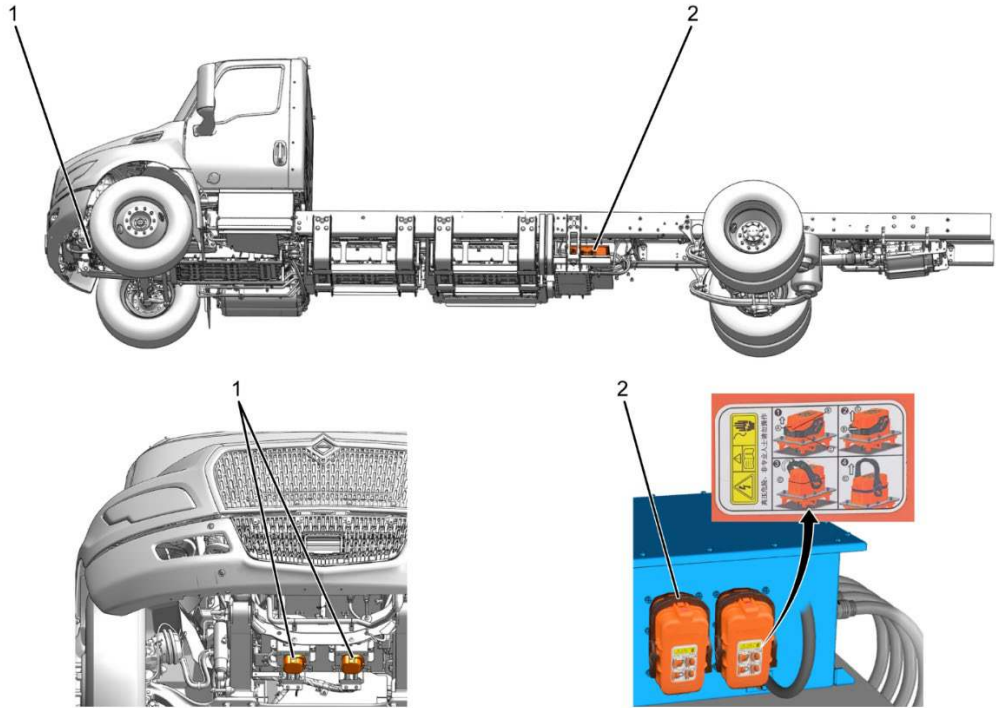


Figure 12. MSD Locations

1. Level 2 MSDs
2. Level 1 MSDs

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, WEAR AND USE APPROVED HIGH-VOLTAGE PERSONAL PROTECTIVE EQUIPMENT (PPE) WHEN NEAR A HIGH-VOLTAGE ELECTRIC VEHICLE. INSPECT PPE BEFORE USE. DO NOT USE GLOVES OR OTHER PPE WITH EXPIRED DATES, HOLES, CRACKS, OR DAMAGE. NEVER TOUCH ENERGIZED ORANGE HIGH-VOLTAGE CABLES OR HIGH-VOLTAGE COMPONENTS WITHOUT WEARING APPROVED HIGH VOLTAGE PPE.

There are two sets of Manual Service Disconnects (MSDs) on the vehicle. The MSDs shut down power to the BLUE-striped cables (Level 2 / Zone 2) and remaining ORANGE cables that were not already shut down by the high-voltage disconnect. WHITE-striped cables (Level 3 / Zone 3) will remain powered after removing the Levels 1 and 2 MSDs.

Inform the tow truck driver of the MSD locations. MSDs typically should not need to be disconnected unless service access is needed near the ORANGE high-voltage cables.

3. Disable Direct Hazards / Safety Regulations

If removing the MSDs, remove them in the following order:

1. The Level 1 MSDs (Figure 12, Item 2), located on the S-box between the frame rails forward of the rear axle.
2. The Level 2 MSDs (Figure 12, Item 1), located on the underside forward of the vehicle.

NOTE: Power still exists in Level 3 / Zone 3 cables (WHITE striped cables) after MSDs are removed.

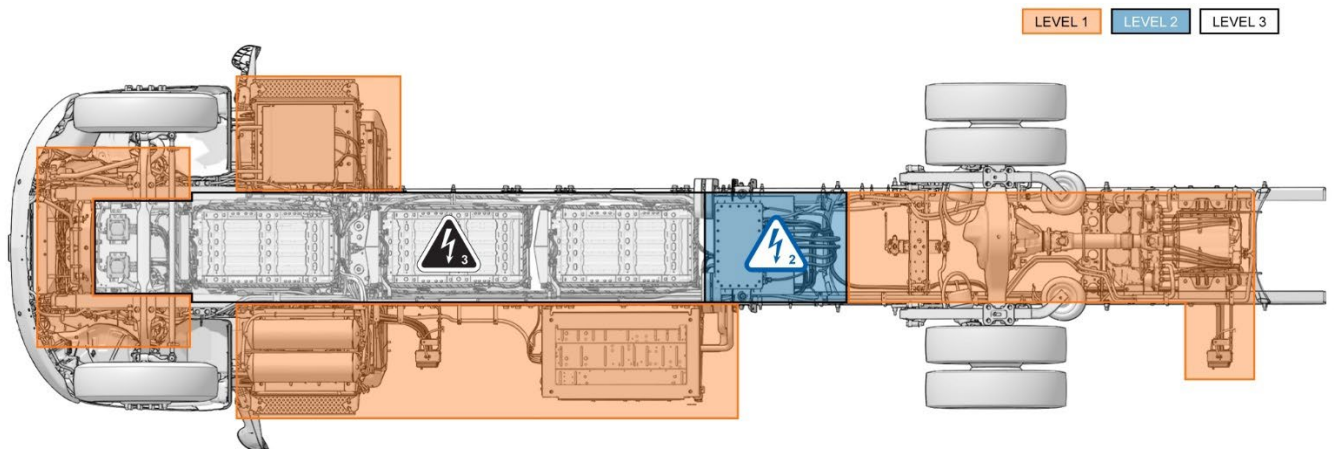


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Figure 13. Level 1 MSDs

Figure 14. Level 2 MSDs



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Figure 15. High-Voltage Isolation Levels / Zones (Viewed From Bottom)

4. Access to the Occupants

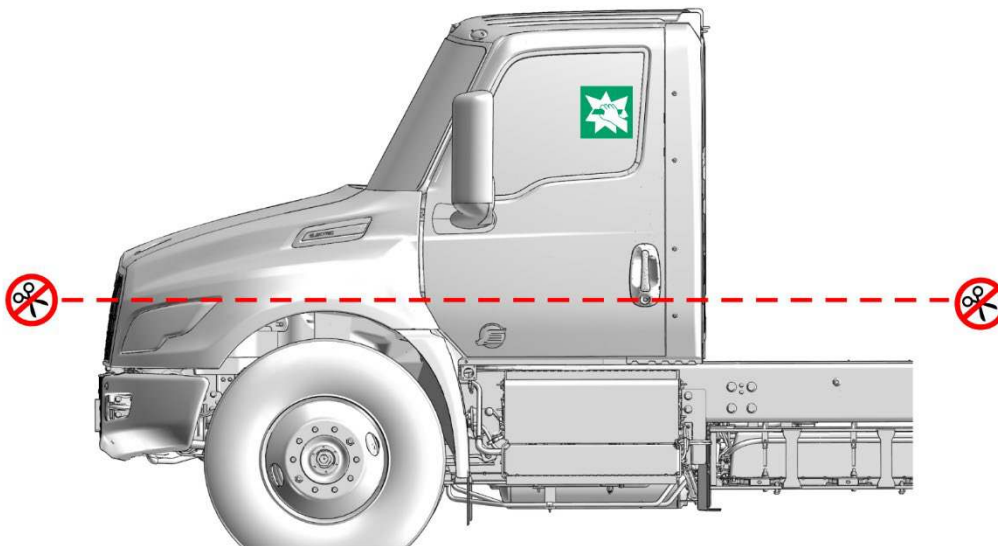
Occupants can be accessed through either the driver or passenger doors located at the front of the vehicle. If equipped, the steering column will tilt and telescope. If equipped, the driver seat can be adjusted by air-operated switches and / or mechanical lever.



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Figure 16. Driver Seat and Steering Column Controls

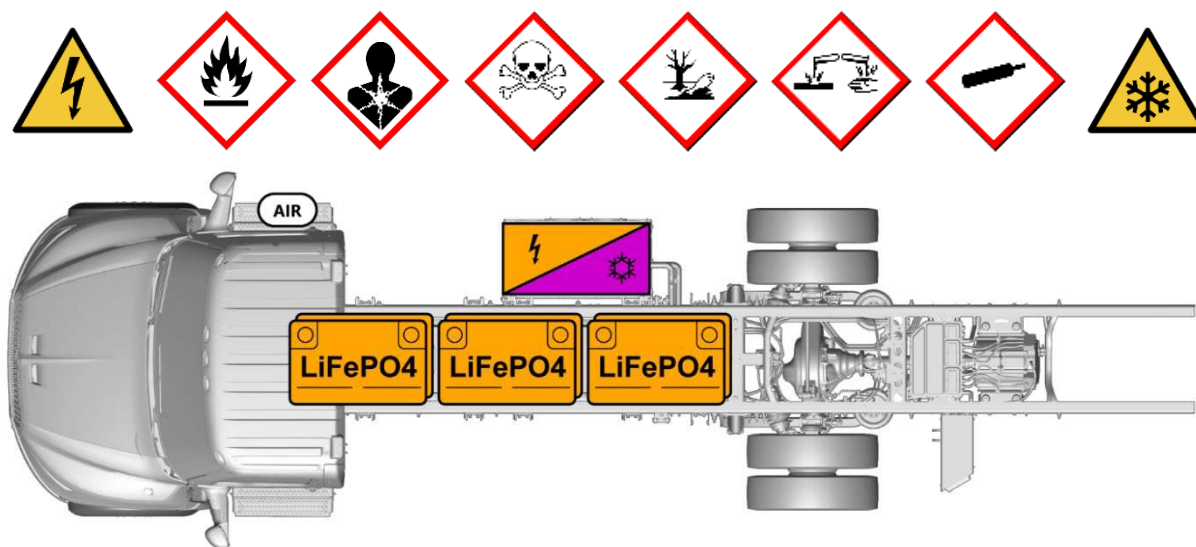
If occupants cannot be accessed through either the driver or passenger doors and cutting is required, do not cut below exterior door handle. The side and rear windows are tempered glass and may be broken for access if needed.



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Figure 17. Do Not Cut Below Line

5. Stored Energy / Liquids / Gases / Solids



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Figure 18. Stored Energy, Liquids, and Gases

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS WEAR FULL PERSONAL PROTECTION EQUIPMENT (PPE), INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA), WHENEVER FIRE IS INVOLVED.

The high-voltage system has six lithium iron phosphate (LiFePO₄) dry cell batteries which are mounted between the frame rails. The high-voltage battery system and cables have a nominal voltage of 609V DC with a maximum potential voltage of 690V DC. These cells are dry cells. Most of the electrolyte stored in these batteries is absorbed by active materials. There is only a small amount of fluid, which may leak if damaged. Leaking electrolyte will occur in drops, not puddles. The fluid is clear in color with a sweet ether-type odor. The smell may indicate a battery leak.

Some of the high-voltage components are liquid cooled with a glycol-based automotive coolant. This liquid is RED in color and may leak in case of a damaged cooling system. The Battery Thermal Management System (BTMS) includes R134a refrigerant to chill the coolant.

The air tanks and system hoses have air pressure of 100–140 psi (690–965 kPa).

Hazardous Material Emergency Cleanup

Following an electric vehicle accident / incident, hazardous materials will need to be cleaned up appropriately.

Please contact your local and state authorities for more information regarding proper response and cleanup of hazardous materials.

6. In Case of Fire



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS WEAR FULL PERSONAL PROTECTION EQUIPMENT (PPE), INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA), WHENEVER FIRE IS INVOLVED. FIRES IN CRASH-DAMAGED ELECTRIC VEHICLES COULD EMIT TOXIC OR COMBUSTIBLE GASES. SMALL AMOUNTS OF EYE, SKIN, OR LUNG IRRITANTS MAY BE PRESENT. IF EXPOSED, RINSE WITH LARGE AMOUNTS OF WATER FOR 10–15 MINUTES. CONSIDER THE ENTIRE VEHICLE AS ENERGIZED.

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. THERE IS A HIGH RISK OF REIGNITION AFTER FIRE IS EXTINGUISHED.

Flames, smoke, arcing, or hot spots like melted plastic may indicate fire or the presence of high voltage. A thermal camera or Infrared (IR) temperature probe may be useful to identify hot spots.

If signs of a fire are present at the scene, follow instructions below:

1. Clear the area around the vehicle if possible. If necessary, move the vehicle to a safe and open space. Keep the vehicle and all combustible materials at least 50 ft (15 m) apart to prevent any potential fire spread.
2. If possible, open the vehicle doors to avoid build-up of gases in the passenger compartment.
3. Identify hot spots with a thermal imaging camera.

NOTE: Refer to and follow state, municipal, and department guidelines for properly handling electric vehicle battery fires.

4. Water may be applied to heat sources to cool or extinguish fires. If vehicle has started burning excessively, it may be more practical to let the battery burn to completion.

Batteries must be completely cooled down before releasing the vehicle to a second responder. Advise second responder that there is always a risk of reignition.

7. In Case of Submersion



In case of submersion, secondary impact damage can never be excluded. Damaged high-voltage components pose an increased electrical shock hazard. Stay away from damaged high-voltage components. Handle any fully or partially submerged vehicle while wearing the appropriate Personal Protection Equipment (PPE).

A submerged vehicle **without** impact damage has a low electrical shock hazard risk. Small bubbles may be noticed exiting the vehicle. This is due to electrolysis of the water and does not create a higher risk of shock hazard.

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. THERE IS A HIGH RISK OF REIGNITION DUE TO DAMAGE AND CORROSION. SALTWATER INCREASES THIS RISK FOR ELECTRICAL SHORTS POST INCIDENT. KEEP FULL PERSONAL PROTECTION EQUIPMENT (PPE) READY, INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA).

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, HANDLE A SUBMERGED VEHICLE WITH APPROPRIATE PERSONAL PROTECTION EQUIPMENT (PPE). CONSIDER THE ENTIRE VEHICLE AS ENERGIZED.

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, AVOID ANY CONTACT WITH A SUBMERGED HIGH-VOLTAGE SYSTEM. DO NOT ATTEMPT TO DISABLE THE HIGH-VOLTAGE SERVICE DISCONNECT SWITCH WHILE THE VEHICLE IS SUBMERGED. THE VEHICLE KEY MAY BE TURNED TO THE OFF POSITION.

In case of submersion, follow instructions below:

1. Turn the vehicle OFF (if possible).
2. Recover the vehicle. Refer to Section 2 – Immobilization / Stabilization / Lifting.
3. Drain the water out of the vehicle.
4. Isolate high-voltage energy. Turn the 12V and High-Voltage Service Disconnect switches to the OFF position.
5. Remove all MSDs. Refer to Section 3 – Disable Direct Hazards / Safety Regulations.

8. Towing / Transportation / Storage



Towing

The preferred method is to tow the vehicle with the rear wheels suspended. If towed with the rear wheels on the ground, the drive shaft / propeller shaft must be removed between the drive motor and rear axle.

The vehicle may be slowly (maximum of 5 mph [8 km/h]) moved anytime over a very short distance to clear the vehicle from traffic. The electric drive motor is connected to the rear wheels and generates electricity for the electric high-voltage system and represents a potential hazard.

The vehicle defaults to Neutral when turned off. Release the parking brake before towing. If the air system has lost pressure, recharge the air tanks on the right side of the vehicle to 64 psi (441 kPa) to provide pressure to release the parking brake before towing.

Transportation

Inform the tow truck driver that reignition is possible at any time with a damaged high-voltage system. If possible, use a thermal or infrared camera to check whether the battery stacks are cold before towing the truck. To ensure that the electric motor is not rotating, remove the drive shaft / propeller shaft before towing the vehicle to the next location.

Storage

Damaged vehicles should be isolated outdoors until inspected. Open windows and doors during isolation to avoid build-up of gases in the compartment. Separate the vehicle from all combustibles, other vehicles, and structures by a distance of at least 50 ft (15 m).

WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. EVEN DAYS LATER SECONDARY FIRE EVENTS CANNOT BE EXCLUDED.

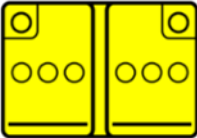
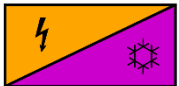
Inform tow truck driver high voltage may still be present in high-voltage batteries and cables.

9. Important Additional Information

Additional information about accident assistance and recovery of vehicle with high-voltage systems can be found at:

- NFPA Alternative Fuel Vehicle Training
<http://www.nfpa.org/for-professionals/training-for-me/buy-alternative-fuel-vehicles-training>
- VDIK Accident Assistance and Recovery of Vehicles with High-Voltage or 48-Volt Systems
vdik.de/wp-content/uploads/2020/07/Accident_Assistance_Recovery_FAQ_en_072020_VDIK-1.pdf
- SAE J2990 Hybrid and EV First and Second Responder Recommended Practice
www.sae.org/standards/content/j2990
- National Transportation Safety Board (NTSB)
www.nts.gov

10. Explanation of Pictograms Used

Pictogram	Designation
	Electric Vehicle
	Battery, Low-Voltage
	Lithium Iron Phosphate Battery
	High-Voltage Component
	Charge Port
	Optional Charge Port
	Battery Thermal Management System

	Device To Shut Down Power in Vehicle
	High-Voltage Disconnect
	12V Shutoff for High-Voltage
	Height Control Truck, by Air System
	High-Voltage Power Cable
	Air Tank
	Break for Access

	Warning, Electricity
	Use Only These Lifting Points
	Hood Release
	Optional Feature
	Optional Tow Hook
	No Cut Below Line
	Flammable
	Hazardous to the Human Health

	Acute Toxicity
	Environmental Hazard
	Corrosives
	Gases Under Pressure
	Warning Low Temperature
	Explosive
	Use Water to Extinguish the Fire
	Use Thermal Infrared Camera

